

CS105.3 Faculty of Computing, Online Examinations 2021

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MODULE NAME (As per the paper)	Database Management Systems		
MODULE CODE	CS105.3		
MODULE LECTURER	Ms. Manoja Weerasekara	DATE SUBMITTED	30/10/ 2021

For office purpose only:

GRADE/MARK	
COMMENTS	

Declaration

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Question 1

a) It is not mandatory.

Database is like a container where we can use to store and manage data. It is data repository. If we can manage it by using a text file or spreadsheets, Don't need to go for a data base. But, when the business is growing or when we get more and more data or when data using by multiple persons, Can be little bit difficult to manage. At that moment, it is necessary to have database.

b) It is not mandatory

DBMS is a software application which helps you to access, manipulate the data that we have stored in the data base. So if we have a small business, then no need to go for a DBMS. It can manage without a database.

c)

i.

- 10 Records.
- 9 Fields per Record.

- ii. City is a part of the address. There is no direct way to extract city details.

Solution :-

Create multiple columns or if not we can add a new column. Then break the address into pieces.

d) 1. S_ID

2. 0NF

3.

4.

0NF

S_ID	M_ID	S_Name	M_Name	Grade	D_ID	D_name	D_Phone

1NF

Student

S_ID	M_ID	S_Name	M_Name	Grade

Department

S_ID	D_ID	D_name	D_Phone

2NF

Student

S_ID	M_ID	S_Name	M_Name	Grade

S_ID	D_ID

Department

D_ID	S_Name	D_Phone

3NF

Students

S_ID	M_ID	S_Name	M_Name	Grade

S_ID	D_ID

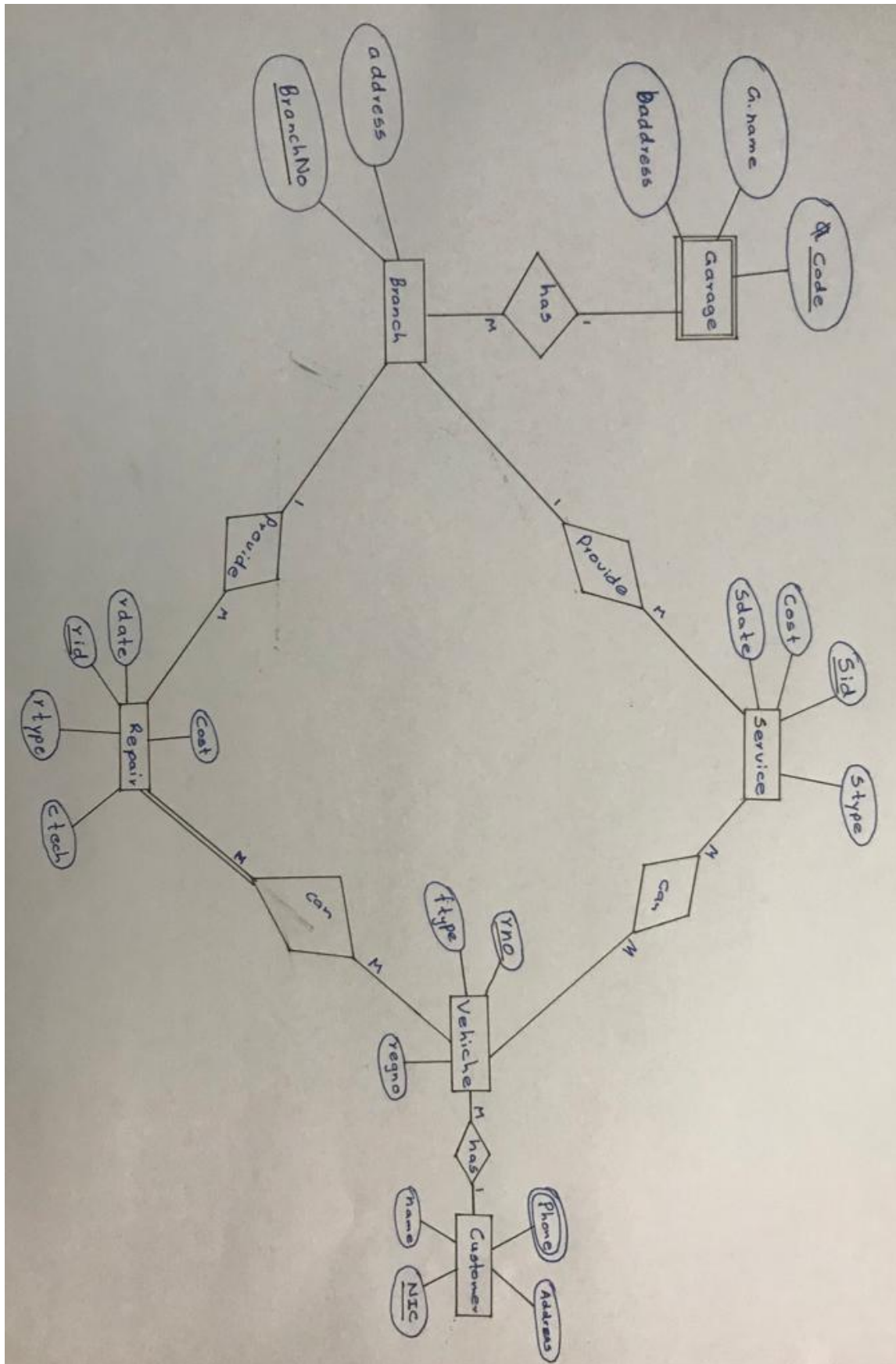
Department

D_ID	S_Name	D_Phone

Module

M_ID	M_Name

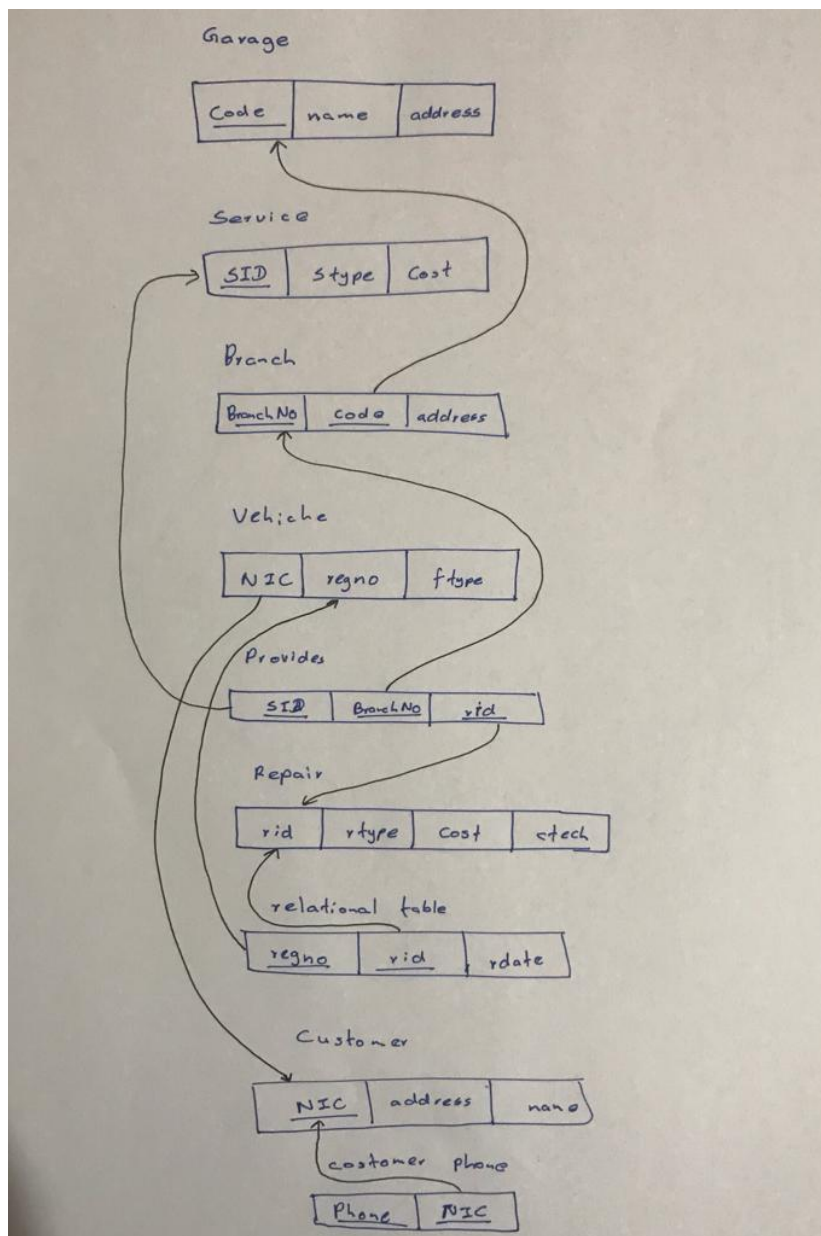
Question 2



a)

No of. Entities identified	06
No.of 1:1 relationships	00
No.of 1:M /M:1 relationships	04
No.of N: M relationships	02

b)



Question 3

a.

```
MariaDB [(none)]> create database 23781_Company;
```

```
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [(none)]> use 23781_Company;
```

```
Database changed
```

```
MariaDB [23781_Company]> create table Department(
```

```
    -> DepID varchar(20),
```

```
    -> DName varchar(30) not null,
```

```
    -> Dphone int not null,
```

```
    -> DLocation varchar(30) not null,
```

```
    -> primary key(DepID));
```

```
Query OK, 0 rows affected (0.634 sec)
```

```
MariaDB [23781_Company]> create table Employee(
```

```
    -> staffID int,
```

```
    -> fName varchar(30) not null,
```

```
    -> LName varchar(30) not null,
```

```
    -> position varchar(50) not null,
```

```
    -> Gender char(1) not null,
```

```
    -> DOB date not null,
```

```
    -> monthlySalary double(10,2) not null,
```

```
    -> DepID varchar(30),
```

```
    -> foreign key(DepID)references Department(DepID));
```

```
Query OK, 0 rows affected (0.411 sec)
```

b.

```
MariaDB [23781_Company]> insert into Department values("C001","Information
System","5445012","Homagama"),("C002","Computer
Science","5445011","Nugegoda"),("C003","Data Science","5445013","Homagama");
Query OK, 3 rows affected (0.143 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
insert into Employee values("101","Manoj","Herath","Senior Lecture","F","1-oct-
86","50000.00","C001"),("102","john","iman","Lecturer","M","3-sep-
96","40000.00","C002"),("201","Amanda","Perera","Registrar","F","12-jan-
75","30000.00","C001"),("103","Amando","dias","Lecturer","F","25-Feb-
98","20000.00","C002");
```

```
Query OK, 4 rows affected, 4 warnings (0.122 sec)
Records: 4 Duplicates: 0 Warnings: 4
```

c.

```
MariaDB [23781_Company]> insert into Employee values("104","vindi","Madly","junior
Lecture","F","25-Feb-99","22000.00","C004");
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`23781_company`.`employee`, CONSTRAINT `employee_ibfk_1` FOREIGN KEY
(`DepID`) REFERENCES `department` (`DepID`))
```

d.

```
MariaDB [23781_Company]> select staffID,fName,LName,MonthlySalary
-> from Employee;
+-----+-----+-----+-----+
| staffID | fName | LName | MonthlySalary |
+-----+-----+-----+-----+
| 101 | Manoj | Herath | 50000.00 |
```


102	john	iman	40000.00
201	Amanda	Perera	30000.00
103	Amando	dias	20000.00

4 rows in set (0.001 sec)

e.

```
MariaDB [23781_Company]> select StaffID,MonthlySalary,(MonthlySalary*12)AS
"AnnualSalary"
```

```
-> from Employee;
```

StaffID	MonthlySalary	AnnualSalary
101	50000.00	600000.00
102	40000.00	480000.00
201	30000.00	360000.00
103	20000.00	240000.00

4 rows in set (0.039 sec)

f.

```
MariaDB [23781_Company]> select Position
```

```
-> from Employee;
```

Position
Senior Lecture
Lecturer
Registrar
Lecturer

4 rows in set (0.001 sec)

g.

MariaDB [23781_Company]> select StaffID

-> from employee

-> where MonthlySalary > 22000.00;

+-----+

| StaffID |

+-----+

| 101 |

| 102 |

| 201 |

+-----+

3 rows in set (0.054 sec)

h.

MariaDB [23781_Company]> select DepID,DPhone

-> from Department

-> where DLocation = "Homagama" OR DLocation = "Galle";

+-----+-----+

| DepID | DPhone |

+-----+-----+

| C001 | 5445012 |

| C003 | 5445013 |

+-----+-----+

2 rows in set (0.016 sec)

i.

```
MariaDB [23781_Company]> select StaffID,fName,1Name  
-> from Employee  
-> where MonthlySalary between 22000.00 and 35000.00;
```

```
+-----+-----+-----+  
| StaffID | fName | 1Name |  
+-----+-----+-----+  
|    201 | Amanda | Perera |  
+-----+-----+-----+  
1 row in set (0.036 sec)
```

j.

```
MariaDB [23781_Company]> select *  
-> from employee  
-> where position = "Senior Lecture" OR Position = "Lecturer";
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+  
| staffID | fName | 1Name | position    | Gender | DOB      | monthlySalary | DepID |  
+-----+-----+-----+-----+-----+-----+-----+-----+  
|    101 | Manoj | Herath | Senior Lecture | F      | 0000-00-00 | 50000.00 | C001 |  
|    102 | john  | iman  | Lecturer     | M      | 0000-00-00 | 40000.00 | C002 |  
|    103 | Amando | dias  | Lecturer     | F      | 0000-00-00 | 20000.00 | C002 |  
+-----+-----+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.000 sec)
```

k.

```
MariaDB [23781_Company]> select DepID
```

```
-> from department
```

```
-> where DLocation = "Nugegoda";
```

```
+-----+
```

```
| DepID |
```

```
+-----+
```

```
| C002 |
```

```
+-----+
```

```
1 row in set (0.001 sec)
```

l.

```
MariaDB [23781_Company]> select*
```

```
-> from Employee
```

```
-> order by MonthlySalary asc;
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
| staffID | fName | lName | position | Gender | DOB | monthlySalary | DepID |
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
| 103 | Amando | dias | Lecturer | F | 0000-00-00 | 20000.00 | C002 |
```

```
| 201 | Amanda | Perera | Registrar | F | 0000-00-00 | 30000.00 | C001 |
```

```
| 102 | john | iman | Lecturer | M | 0000-00-00 | 40000.00 | C002 |
```

```
| 101 | Manoj | Herath | Senior Lecture | F | 0000-00-00 | 50000.00 | C001 |
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
4 rows in set (0.028 sec)
```